Quantitative Methods – Human Sciences

Reading list

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For every topic covered in this class, I am borrowing references from across several disciplines (mostly Sociology, Economics, Political Science, Epidemiology, Statistics, and Computer Sciences). Causal inference is an interdisciplinary effort, and no single discipline covers all fronts. Additionally, different disciplines have slightly different ways of explaining and justifying things, and you may find some of them more helpful and illuminating than others. Eventually, I hope you will develop the skill to read across these niches and fill in the gaps when background knowledge is assumed.

There is another reason why I believe it is important to broaden the spectrum of one's readings. When doing research, you will encounter cases in which the research designs you already know are not really suitable for your application. Lacking good examples, you may be tempted to force your analysis into something known. Or, if you're very committed, you're going to find yourself reinventing the wheel. But it may well be the case that the answer to your question has been found somewhere else! You're just not going to find it if you only search inside your disciplinary journals.

That said, the only *required* readings are from Huntington-Klein's *The Effect* (2nd ed., 2025), which is available online for free [link]. The remaining references provide further context, including the historical genealogy of the concepts we are using, or go more in depth into some aspect of the problem, including some cutting-edge developments. Feel free to explore them as you see fit. It is fine if you don't get everything in the first pass (in fact, it is expected). It is also fine if you need to pause, read again, find a Youtube video explaining the math, bring some paper and a pen, and work through some details yourself.

1 Required readings

Week 1: Causality and the experimental ideal

The Effect, chapters 1-4

Week 2: A crash course on Directed Acyclic Graphs

The Effect, chapters 5-8

Week 3: Selection on observables I (matching, regression, weighting)

The Effect, chapters 13-14

Week 4: Selection on observables II (matching, regression, weighting)

The Effect, chapters 13-14

Week 5: As-if-random assignment (instrumental variables and quasi-experiments)

The Effect, chapters 19-20

Week 6: Parallel trends I (difference-in-differences and extensions)

The Effect, chapters 16-18

Week 7: Parallel trends II (difference-in-differences and extensions)

The Effect, chapters 16-18

Week 8: Sensitivity analysis (quantitative bias analysis)

The Effect, chapter 21

2 Further readings

2.1 Why causal inference?

***Keywords**: theoretical estimand, empirical estimand, estimator, estimate, assumptions*

Start here

Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. 2021. "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." American Sociological Review 86(3):532–65. doi: 10.1177/00031224211004187. (**)

Morgan, Stephen L. and Christopher Winship (2014). Counterfactuals and Causal Inference: Methods and Principles for Social Research (2nd edition), Cambridge University Press, Chapters 1-2 (*)

If you want more

Greenland, Sander. 2022. "The Causal Foundations of Applied Probability and Statistics." Pp. 605–24 in Probabilistic and Causal Inference: The Works of Judea Pearl. Vol. 36. New York, NY, USA: Association for Computing Machinery.

Hernán, Miguel A., John Hsu, and Brian Healy. 2019. "A Second Chance to Get Causal Inference Right: A Classification of Data Science Tasks." CHANCE 32(1):42–49. doi: 10.1080/09332480.2019.1579578.

Imbens, Guido W. 2022. "Causality in Econometrics: Choice vs Chance." Econometrica 90(6):2541–66. doi: 10.3982/ECTA21204.

Mitra, Nandita, Jason Roy, and Dylan Small. 2022. "The Future of Causal Inference." American Journal of Epidemiology 191(10):1671–76. doi: 10.1093/aje/kwac108.

Ogburn, Elizabeth L., and Ilya Shpitser. 2021. "Causal Modelling: The Two Cultures." Observational Studies 7(1):179–83.

Pearl, Judea, and Dana Mackenzie. 2019. The Book of Why: The New Science of Cause and Effect. 1st edition. London: Penguin.

2.2 Potential outcomes to hypothesize, experiments to learn

***Keywords**: potential outcomes, switching equation, treatment effect, SUTVA, consistency, exchangeability, ignorability, independence, identification, randomization, balance, clusters, blocks*

Start here

Gerber, Alan S., and Donald P. Green. 2012. Field Experiments – Design, Analysis, and Interpretation. Illustrated edition. New York London: W. W. Norton & Company, Chapters 1-3 (*)

Holland, Paul W. 1986. "Statistics and Causal Inference." Journal of the American Statistical Association 81(396):945–60. doi: 10.2307/2289064.

Pager, Devah. 2003. "The Mark of a Criminal Record." American Journal of Sociology 108(5):937–75. doi: 10.1086/374403.

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If you want more

Deaton, Angus, and Nancy Cartwright. 2018. "Understanding and Misunderstanding Randomized Controlled Trials." Social Science & Medicine 210:2–21. doi: 10.1016/j.socscimed.2017.12.005.

Doan, Long, Natasha Quadlin, and Katharine Khanna. 2024. "Using Experiments to Study Families and Intimate Relationships." Journal of Marriage and Family 86(5):1251–71. doi: 10.1111/jomf.12959.

Druckman, James N. 2022. Experimental Thinking: A Primer on Social Science Experiments. Cambridge: Cambridge University Press.

Elwert, Felix, Tamás Keller, and Andreas Kotsadam. 2023. "Rearranging the Desk Chairs: A Large Randomized Field Experiment on the Effects of Close Contact on Interethnic Relations." American Journal of Sociology 128(6):1809–40. doi: 10.1086/724865.

Glennerster, Rachel, and Kudzai Takavarasha. 2013. Running Randomized Evaluations: A Practical Guide. Illustrated edition. Princeton, NJ: Princeton University Press.

Imbens, Guido W., and Donald B. Rubin. 2015. Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction. Cambridge: Cambridge University Press, Chapters 1-3

Kohavi, Ron. 2020. Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing. 1st edition. Cambridge: Cambridge University Press.

Pager, Devah, Rebecca Goldstein, Helen Ho, and Bruce Western. 2022. "Criminalizing Poverty: The Consequences of Court Fees in a Randomized Experiment." American Sociological Review 87(3):529–53. doi: 10.1177/00031224221075783.

Quadlin, Natasha. 2018. "The Mark of a Woman's Record: Gender and Academic Performance in Hiring." American Sociological Review 83(2):331–60. doi: 10.1177/0003122418762291.

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Sävje, Fredrik. 2021. "Randomization Does Not Imply Unconfoundedness."

Veltri, Giuseppe. 2023. Designing Online Experiments for the Social Sciences. 1st edition. Los Angeles London New Delhi Singapore Washington DC Melbourne: SAGE Publications Ltd.

Weisshaar, Katherine, Koji Chavez, and Tania Hutt. 2024. "Hiring Discrimination Under Pressures to Diversify: Gender, Race, and Diversity Commodification across Job Transitions in Software Engineering." American Sociological Review 89(3):584–613. doi: 10.1177/00031224241245706.

2.3 Structural causal model as scaffoldings

***Keywords**: structural equations, directed acyclic graphs, chain, fork, collider, do-operator, interventional graph, testing graph, backdoor criterion, frontdoor criterion, adjustment formula, non-parametric identification*

Start here

Elwert, Felix, and Christopher Winship. 2014. "Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable." Annual Review of Sociology 40(1):31–53. doi: 10.1146/annurev-soc-071913-043455. (**)

Morgan, Stephen L. and Christopher Winship (2014). Counterfactuals and Causal Inference: Methods and Principles for Social Research (2nd edition), Cambridge University Press, Chapter 3 (*)

Pearl, Judea, Madelyn Glymour, and Nicholas P. Jewell. 2016. Causal Inference in Statistics: A Primer. 1st edition. Chichester: Wiley, Chapter 3 (until section 3.7)

If you want more

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Hernán, Miguel A., Sonia Hernández-Díaz, and James M. Robins. 2004. "A Structural Approach to Selection Bias" Epidemiology 15(5):615–25. doi: 10.1097/01.ede.0000135174.63482.43.

Imbens, Guido W., and Donald B. Rubin. 1995. "Causal Diagrams for Empirical Research: Discussion of 'Causal Diagrams for Empirical Research' by J. Pearl." Biometrika 82(4):694–95. doi: 10.1093/biomet/82.4.694.

Keele, Luke, Randolph T. Stevenson, and Felix Elwert. 2020. "The Causal Interpretation of Estimated Associations in Regression Models." Political Science Research and Methods 8(1):1–13. doi: 10.1017/psrm.2019.31.

Knight, Carly R., and Christopher Winship. 2013. "The Causal Implications of Mechanistic Thinking: Identification Using Directed Acyclic Graphs (DAGs)." Pp. 275–99 in Handbook of Causal Analysis

for Social Research, Handbooks of Sociology and Social Research, edited by S. L. Morgan. Dordrecht: Springer Netherlands.

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2.4 Identification strategies as research templates

***Keywords**: identification strategies, identification assumptions, credibility ladder, (non-)parametric assumptions, as-if-random, observational studies, quasi-experiments, natural experiments*

Start here

Imbens, Guido W. 2024. "Causal Inference in the Social Sciences." Annual Review of Statistics and Its Application 11(Volume 11, 2024):123–52. doi: 10.1146/annurev-statistics-033121-114601.

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Imbens, Guido W. 2020. "Potential Outcome and Directed Acyclic Graph Approaches to Causality: Relevance for Empirical Practice in Economics." Journal of Economic Literature 58(4):1129–79. doi: 10.1257/jel.20191597.

Lewbel, Arthur. 2019. "The Identification Zoo: Meanings of Identification in Econometrics." Journal of Economic Literature 57(4):835–903. doi: 10.1257/jel.20181361.

Steiner, Peter M., Yongnam Kim, Courtney E. Hall, and Dan Su. 2017. "Graphical Models for Quasi-Experimental Designs." Sociological Methods & Research 46(2):155–88. doi: 10.1177/0049124115582272.

Winship, Christopher, and Stephen L. Morgan. 1999. "The Estimation of Causal Effects from Observational Data." Annual Review of Sociology 25(Volume 25, 1999):659–706. doi: 10.1146/annurev.soc.25.1.659.

2.5 Modeling $E(Y|X)$: regression adjustment and imputation

***Keywords**: adjustment set, admissible set, adjustment formula, deconfounding, regression analysis, (partial) linear model, imputation, g-computation*

Start here

Angrist, Joshua D., Jorn-steffen Pischke, and Jörn-steffen Pischke. 2009. Mostly Harmless Econometrics – An Empiricist's Companion. Illustrated edition. Princeton: Princeton University Press, Regression Chapter (*)

Cinelli, Carlos, Andrew Forney, and Judea Pearl. 2024. "A Crash Course in Good and Bad Controls." Sociological Methods & Research 53(3):1071–1104. doi: 10.1177/00491241221099552.(**)

Chatton, Arthur, and Julia M. Rohrer. 2024. "The Causal Cookbook: Recipes for Propensity Scores, G-Computation, and Doubly Robust Standardization." Advances in Methods and Practices in Psychological Science 7(1). doi: 10.1177/25152459241236149.

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Goldsmith-Pinkham, Paul, Peter Hull, and Michal Kolesár. 2024. "Contamination Bias in Linear Regressions." American Economic Review 114(12):4015–51. doi: 10.1257/aer.20221116.

Hainmueller, Jens, and Chad Hazlett. 2014. "Kernel Regularized Least Squares: Reducing Misspecification Bias with a Flexible and Interpretable Machine Learning Approach." Political Analysis 22(2):143–68. doi: 10.1093/pan/mpt019.

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2.6 Modeling $E(D|X)$: matching and weighting

***Keywords**: propensity score, greedy matching, optimal matching, inverse probability weighting, calibration weighting, balance in expectation, balance in sample*

Start here

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Zhou, Xiang. 2019. "Equalization or Selection? Reassessing the 'Meritocratic Power' of a College Degree in Intergenerational Income Mobility." American Sociological Review 84(3):459–85. doi: 10.1177/0003122419844992.

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Hazlett, Chad. 2016. "Kernel Balancing: A Flexible Non-Parametric Weighting Procedure for Estimating Causal Effects." SSRN Electronic Journal. doi: 10.2139/ssrn.2746753.

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Zubizarreta, José R. 2015. "Stable Weights That Balance Covariates for Estimation With Incomplete Outcome Data." Journal of the American Statistical Association 110(511):910–22. doi: 10.1080/01621459.2015.1023805.

Zubizarreta, José R., Ricardo D. Paredes, and Paul R. Rosenbaum. 2014. "Matching for Balance, Pairing for Heterogeneity in an Observational Study of the Effectiveness of for-Profit and Not-for-Profit High Schools in Chile." The Annals of Applied Statistics 8(1):204–31. doi: 10.1214/13-AOAS713.

2.7 Using the side-door: instrumental variables

***Keywords**: instrument, as-if-random, exogeneity, exclusion restriction, monotonicity, homogeneity, principal strata, compliers, local average treatment effect*

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Felton, Chris, and Brandon M. Stewart. 2024. "Handle with Care: A Sociologist's Guide to Causal Inference with Instrumental Variables." Sociological Methods & Research 00491241241235900. doi: 10.1177/00491241241235900. (*)

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Angrist, Joshua D., Guido W. Imbens, and Donald B. Rubin. 1996. "Identification of Causal Effects Using Instrumental Variables." Journal of the American Statistical Association 91(434):444–55. doi: 10.1080/01621459.1996.10476902.

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Legewie, Joscha, and Thomas A. DiPrete. 2012. "School Context and the Gender Gap in Educational Achievement." American Sociological Review 77(3):463–85. doi: 10.1177/0003122412440802.

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Swanson, Sonja A., and Miguel A. Hernán. 2014. "Think Globally, Act Globally: An Epidemiologist's Perspective on Instrumental Variable Estimation." Statistical Science 29(3):371–74.

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2.8 Peeking through the window: regression discontinuity

***Keywords**: running variable, threshold, local effect, extrapolation, local randomization, continuity of potential outcomes, sorting, sharp rdd, fuzzy rdd*

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***Keywords**: parallel trends, no anticipation, two-way fixed effects, event study, absorbing treatment, time-constant unobserved confounding*

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***Keywords**: heterogeneity, staggered adoption, negative weights, forbidden comparison, synthetic control, matrix completion, trajectory balancing*

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